Experiment No. 7

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Roll No : 3025

Batch : T2

Subject : Machine Learning

Title: Weka Tool Report

Introduction

WEKA (Waikato Environment for Knowledge Analysis) is an open-source software developed by the University of Waikato for machine learning and data mining tasks. It provides a collection of visualization tools and algorithms for data analysis and predictive modeling. WEKA is written in Java and is widely used for classification, regression, clustering, and association rule mining.

Download Steps and Version

Go to official website:
WEKA (search on Google: *WEKA download*

Click Download

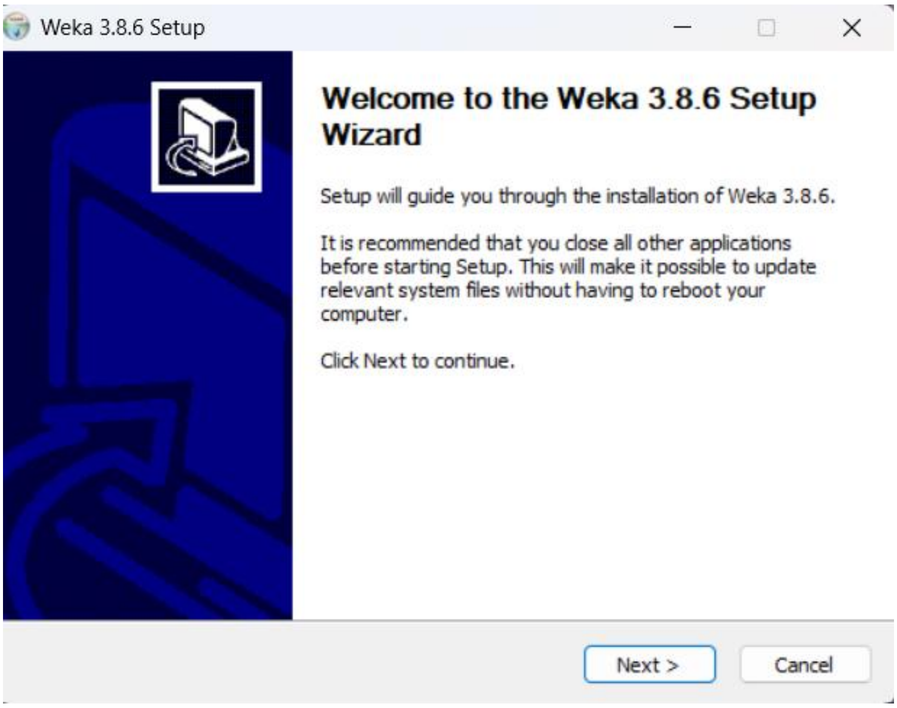
Choose your OS (Windows)

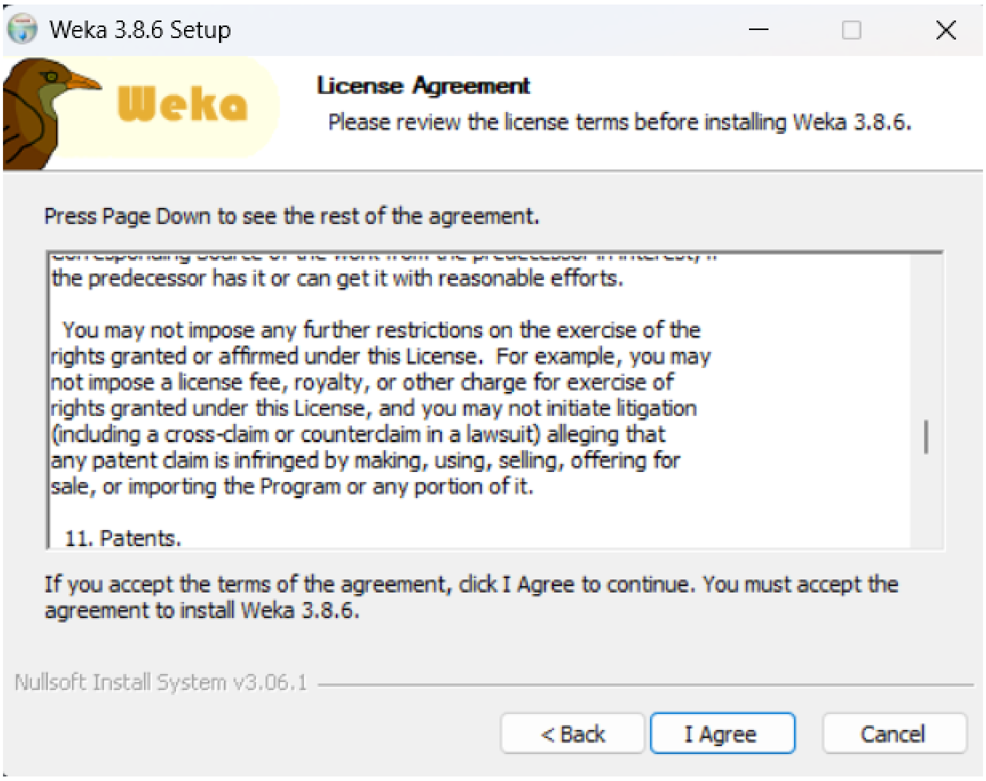
Install like normal software

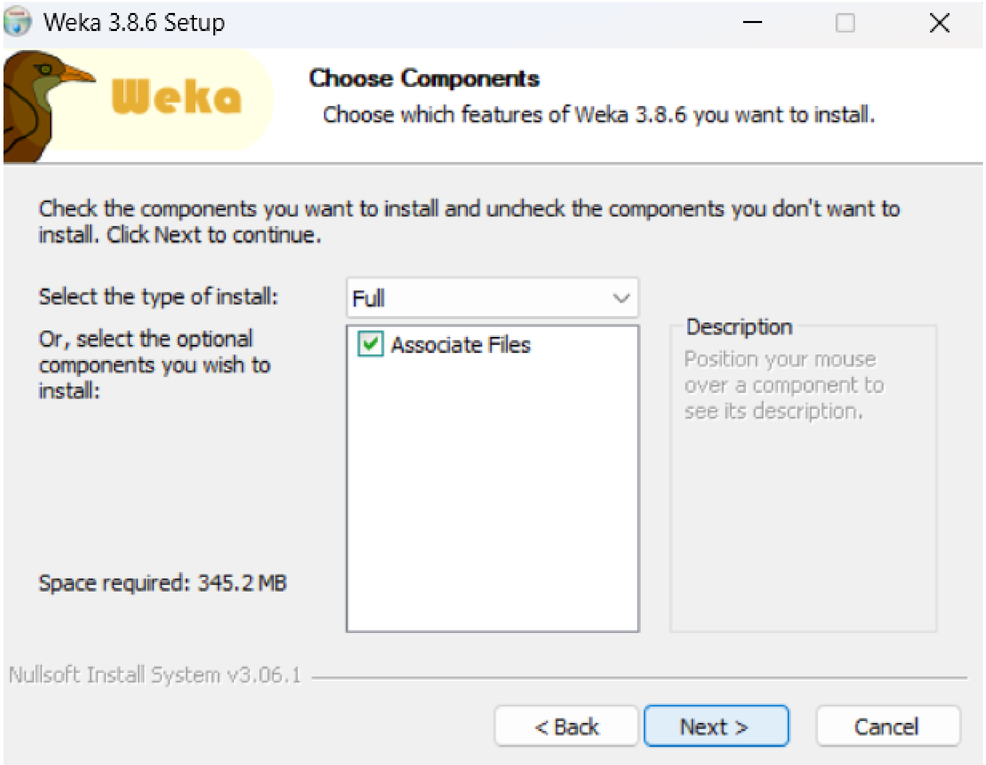
Version :

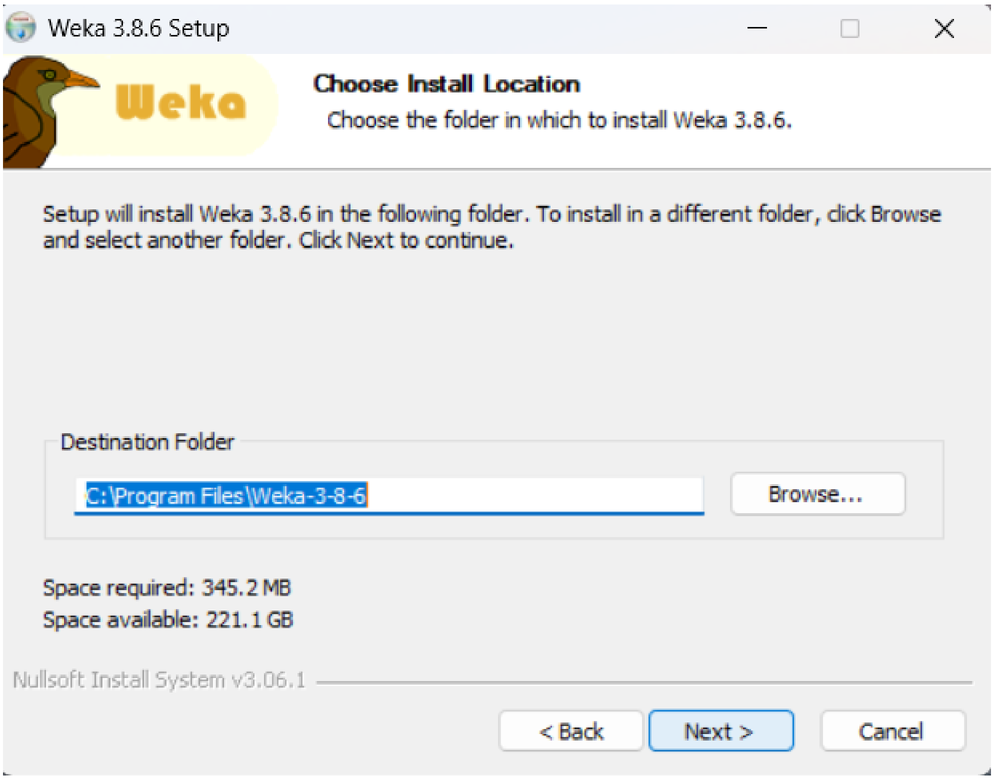
3.8.6

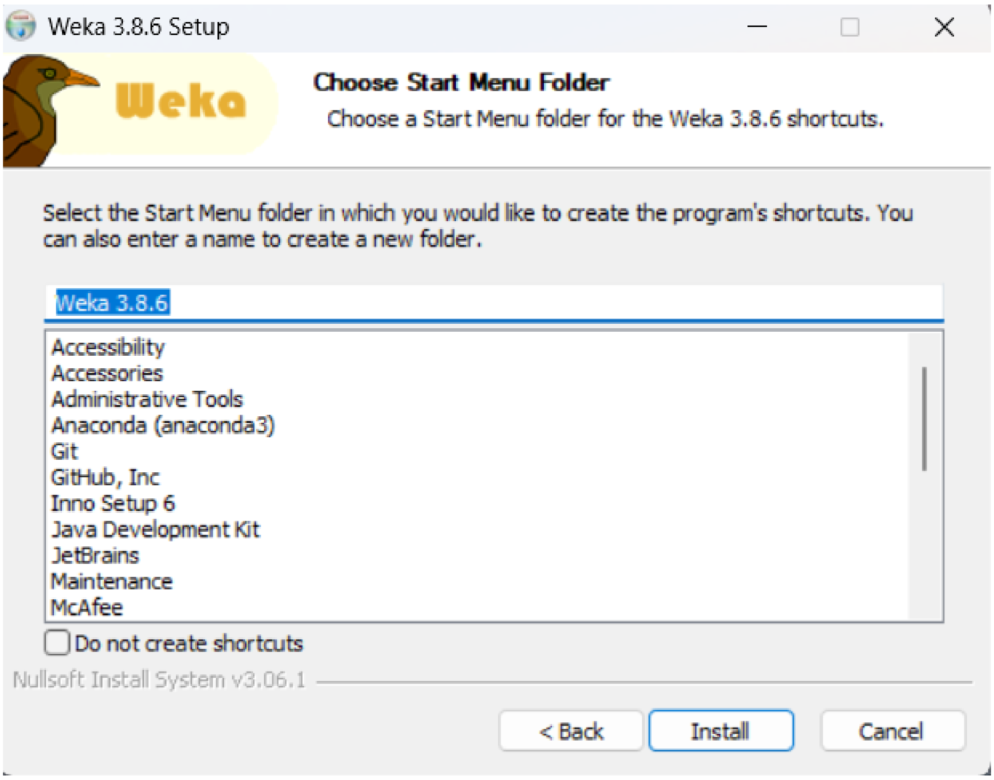
Screenshots:

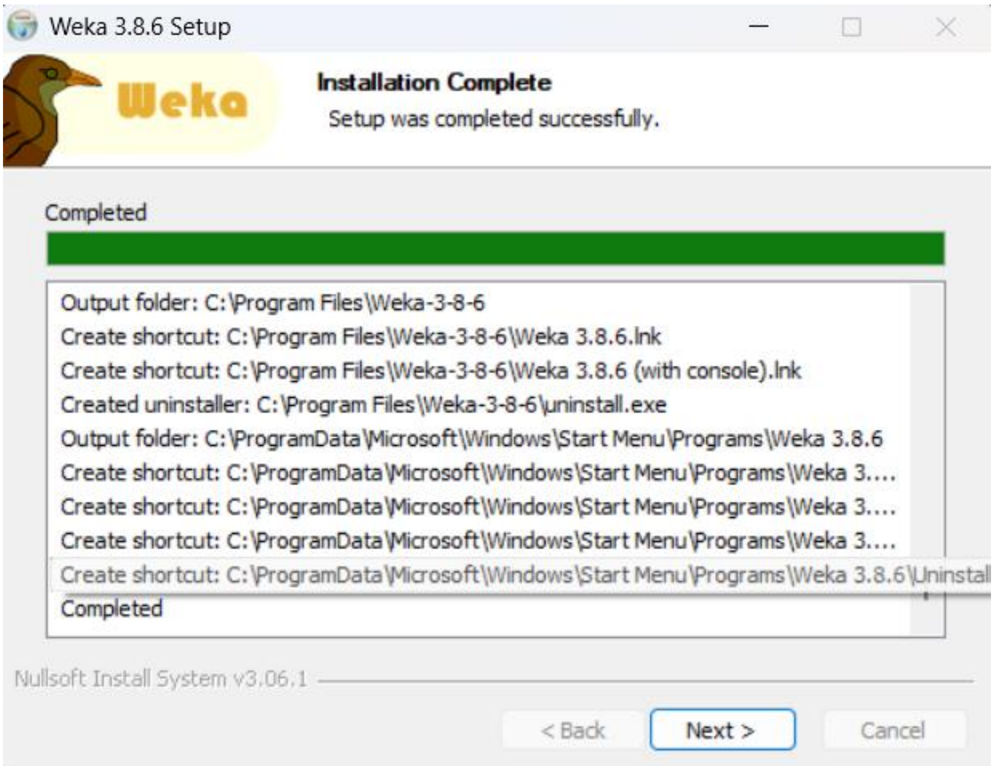

Weka 3.8.6 Setup
Welcome to the Weka 3.8.6 Setup Wizard
Setup will guide you through the installation of Weka 3.8.6.
It is recommended that you close all other applications before starting Setup. This will make it possible to update relevant system files without having to reboot your computer.
Click Next to continue.
Next >
Cancel

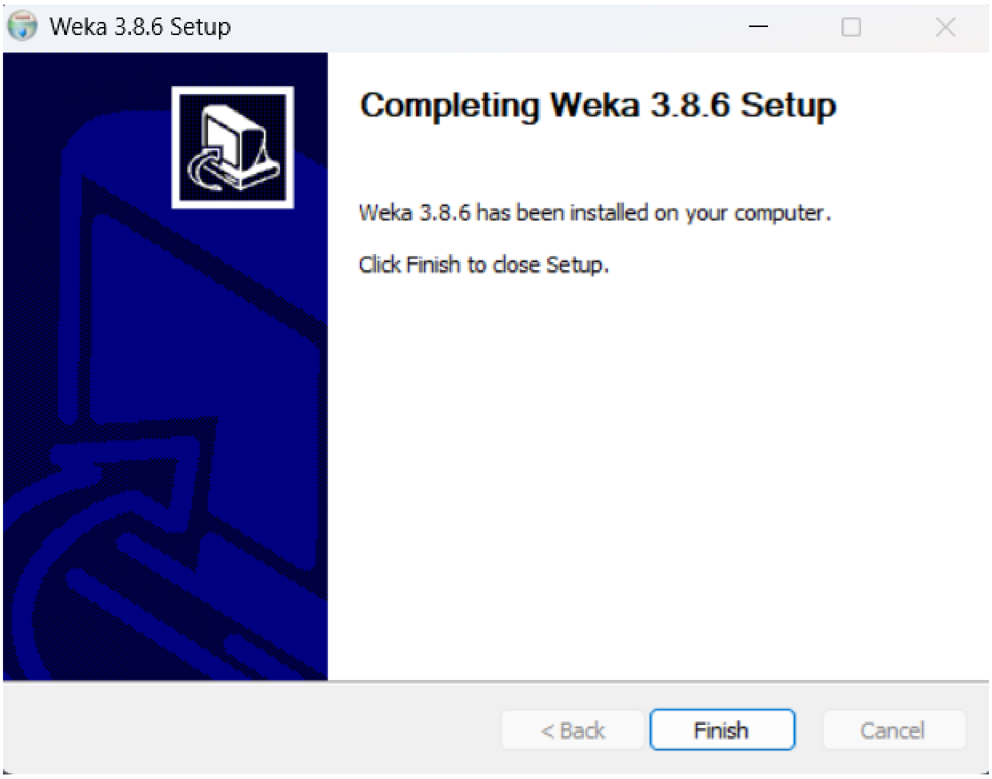

Weka 3.8.6 Setup
Weka
License Agreement
Please review the license terms before installing Weka 3.8.6.
Press Page Down to see the rest of the agreement.
the predecessor has it or can get it with reasonable efforts.
You may not impose any further restrictions on the exercise of the rights granted or affirmed under this License. For example, you may not impose a license fee, royalty, or other charge for exercise of rights granted under this License, and you may not initiate litigation (including a cross-claim or counterclaim in a lawsuit) alleging that any patent claim is infringed by making, using, selling, offering for sale, or importing the Program or any portion of it.
11. Patents.
If you accept the terms of the agreement, click I Agree to continue. You must accept the agreement to install Weka 3.8.6.
Nullsoft Install System v3.06.1
< Back
I Agree
Cancel


Weka 3.8.6 Setup
Weka
Choose Components
Choose which features of Weka 3.8.6 you want to install.
Check the components you want to install and uncheck the components you don't want to install. Click Next to continue.
Select the type of install:
Full
Or, select the optional components you wish to install:
Associate Files
Description
Position your mouse over a component to see its description.
Space required: 345.2 MB
Nullsoft Install System v3.06.1
< Back
Next >
Cancel


Weka 3.8.6 Setup
Weka
Choose Install Location
Choose the folder in which to install Weka 3.8.6.
Setup will install Weka 3.8.6 in the following folder. To install in a different folder, click Browse and select another folder. Click Next to continue.
Destination Folder
C:\Program Files\Weka-3-8-6
Browse...
Space required: 345.2 MB
Space available: 221.1 GB
Nullsoft Install System v3.06.1
< Back
Next >
Cancel


Weka 3.8.6 Setup
Weka
Choose Start Menu Folder
Choose a Start Menu folder for the Weka 3.8.6 shortcuts.
Select the Start Menu folder in which you would like to create the program's shortcuts. You can also enter a name to create a new folder.
Weka 3.8.6
Accessibility
Accessories
Administrative Tools
Anaconda (anaconda3)
Git
GitHub, Inc
Inno Setup 6
Java Development Kit
JetBrains
Maintenance
McAfee
Do not create shortcuts
Nullsoft Install System v3.06.1
< Back
Install
Cancel


Weka 3.8.6 Setup
Weka
Installation Complete
Setup was completed successfully.
Completed
Output folder: C:\Program Files\Weka-3-8-6
Create shortcut: C:\Program Files\Weka-3-8-6\Weka 3.8.6.lnk
Create shortcut: C:\Program Files\Weka-3-8-6\Weka 3.8.6 (with console).lnk
Created uninstaller: C:\Program Files\Weka-3-8-6\uninstall.exe
Output folder: C:\ProgramData\Microsoft\Windows\Start Menu\Programs\Weka 3.8.6
Create shortcut: C:\ProgramData\Microsoft\Windows\Start Menu\Programs\Weka 3....
Create shortcut: C:\ProgramData\Microsoft\Windows\Start Menu\Programs\Weka 3....
Create shortcut: C:\ProgramData\Microsoft\Windows\Start Menu\Programs\Weka 3....
Create shortcut: C:\ProgramData\Microsoft\Windows\Start Menu\Programs\Weka 3.8.6\Uninstall
Completed
Nullsoft Install System v3.06.1
< Back
Next >
Cancel



Benefits

- Easy to use GUI (no coding required)
- Supports many ML algorithms
- Free and open-source
- Data visualization tools available
- Works with CSV and ARFF files
- Good for beginners in Machine Learning

How to use WEKA

Step 1: Open WEKA

Click → Explorer

Step 2: Load Dataset

- Click Open file

- Open .csv file
- Done

Alogrithm Used

Algorithm Used: J48 Decision Tree

In this project, the J48 algorithm was used, which is an implementation of the Decision Tree algorithm. It is used for classification tasks. The algorithm works by splitting the dataset into subsets based on attribute values, forming a tree-like structure. It is simple, efficient, and easy to interpret.

Conclusion

WEKA is a powerful and user-friendly tool for machine learning. It helps beginners understand algorithms without coding. The J48 algorithm provided accurate classification results on the dataset.